

Experiment No. 9

Inorganic Qualitative Analysis (Salt no. 5)

Aim: Analyse one basic (cation) and one acidic (anion) radical from given inorganic salt.

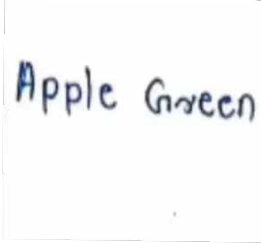
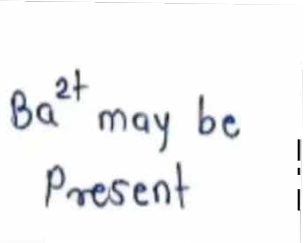
Apparatus: Test tubes, test tube stand, spatula, glass rod, watch glass, beaker, burner, etc.

A. Preliminary tests: (General scheme)

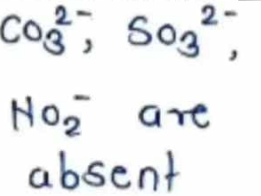
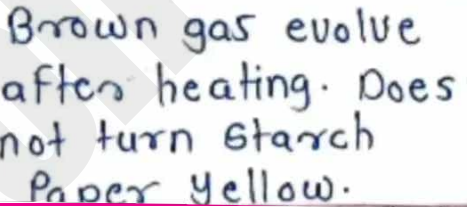
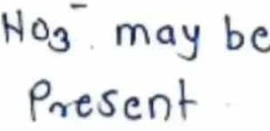
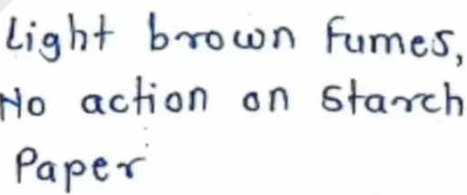
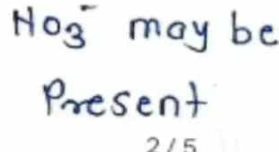
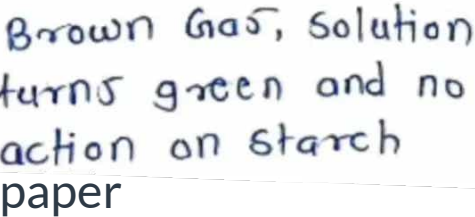
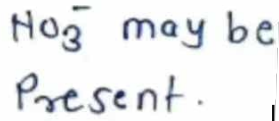
Test	Observation	Inference
Nature	Crystalline (Soluble in water)	Cl^- , Br^- , I^- , NO_3^- , SO_4^{2-} may be Present
Color of salt	white	Pb^{2+} , Al^{3+} , Mg^{2+} , Ba^{2+} , Zn^{2+} , NH_4^+ , Ca^{2+} , Mn^{2+} , Sr^{2+} may be Present
Solubility	soluble in water	crystalline salt Present

B. Dry tests for basic radicals (Cations)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of salt in a clean, dry test tube and heat it strongly. Observe the color of the residue and gas evolved.	white infusible residue	Al^{3+} , Zn^{2+} , Ba^{2+} , Ca^{2+} , NH_4^+ , may be Present
2. Test for NH_4^+ Salt + NaOH, heat gently, hold moist turmeric paper near the mouth of test tube.	No reaction	-
3. Charcoal cavity test: Take the salt and solid Na_2CO_3 in the ratio 1:2 on watch glass, add few drops of water and prepare a paste, place this in charcoal cavity and heat it with a blow pipe in a reducing flame.	white infusible residue	Zn^{2+} , Al^{3+} , Mg^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+} may be Present
4. Cobalt Nitrate Test: If white infusible residue is obtained in charcoal cavity test, add two drops of cobalt nitrate solution on the white residue and heat in the blue flame.	Residue turns grey	Ba^{2+} , Ca^{2+} may be Present.

5. Flame test: Clean a platinum wire with conc. HCl. Moisten a small quantity of the substance with concentrated HCl on watch glass. Dip the cleaned platinum wire in it and heat it in oxidizing flame (non - luminous) and observe the color of the flame.		
6. Borax bead test: [For colored salt only] Make a small loop at the end of a clean platinum wire heat it strongly and dip it in borax powder. Heat it again till a glossy bead is obtained. Now touch the hot bead with a small crystal of the salt and heat it again in oxidizing (blue) flame. Observe the color of the bead.		

C. Dry tests for acidic radicals (Anions)

Test	Observations	Inference
i. Action of dil HCl: Salt + dil HCl.		
ii. Action of conc. H₂SO₄: Salt + conc. H₂SO₄ + heat		
iii. Action of MnO₂ and conc. H₂SO₄: Salt + a pinch of MnO₂ + conc. H₂SO₄ and heat		
iv. Cu-foil test: (If brown gas is obtained in the above test then only perform this test) Salt + Cu-foil + conc. H₂SO₄ and heat strongly		

Note: (If the anion is not detected in above dry test, then salt may contain SO_4^{2-} or PO_4^{3-})

D. Preparation of solution (Original Solution): Dissolve about 0.5 gm of salt in 20 mL of water in a 50 mL beaker. This is called original solution (O.S.)

E. Wet test for basic radical (cations)

Detection of groups

Test	Observations	Inference
1. O.S. + dil. HCl	No ppt	Group I absent
2. O.S. + dil. HCl + H_2S gas/ H_2S water solution	No ppt	Group II absent
3. O.S. + NH_4Cl + NH_4OH till alkaline	No ppt	Group III A absent
4. O.S. + NH_4Cl + NH_4OH till alkaline + H_2S gas/ H_2S water solution	No ppt	Group III B absent
5. O.S. + NH_4Cl + NH_4OH till alkaline + saturated solution of $(\text{NH}_4)_2\text{CO}_3$	white ppt	Group IV ($\text{Ca}^{2+}/\text{Sr}^{2+}/\text{Ba}^{2+}$) may be Present
6. O.S. + NH_4Cl + NH_4OH till alkaline + Na_2HPO_4	No reaction	Group II absent
7. If all the above groups are absent	—	—

Analysis of Group: After detection of a particular group do analysis of that group.

Analysis of Group **IV**

Test	Observation	Inference
O.S + K_2CrO_4	yellow ppt	Ba^{2+} Present
C.T. for... Ba^{2+} 1. O.S + K_2CrO_4	yellow ppt	Ba^{+2} confirmed
2. O.S + dil. H_2SO_4	white ppt	Ba^{+2} confirmed

F. Wet test for acidic radicals:

Detection of acidic radical from water solution:

1. Wet tests for CO_3^{2-} , SO_3^{2-} , Cl^- , Br^- , I^-

1. O.S. + AgNO_3	No ppt	SO_4^{2-} , PO_4^{2-} absent
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Distinction between halides (Cl^- , Br^- , I^-):

Note: (Perform this test only if ppt is obtained for AgNO_3 which is insoluble in dil. HNO_3)

Test	Observation	Inference
O.S. + Chloroform + Cl_2 water in excess (shake well)	No ppt	NO_2^- absent

Note: (If one of the halide radical is present then further analysis is not necessary. Perform confirmatory test for that detected radical.)

2. Wet test for SO_4^{2-} and PO_4^{3-}

(This test is required to perform only, if ppt is NOT obtained in AgNO_3 test)

Test	Observation	Inference
O.S. + $\text{Ba}(\text{NO}_3)_2$	—	—

3. WET TEST FOR NO_2^-

Test	Observation	Inference
O.S. + dil. H_2SO_4 + FeSO_4 solution	Evolution of NO_2 gas which turns starch iodide paper black	NO_3^- Present

4. WET TEST FOR NO_3^-

Test	Observation	Inference
O.S. + 3-4 drops of conc. H_2SO_4 + 2-3 drops of diphenyl amine	Evolution of NO_2 gas which turns starch iodide paper black	NO_3^- Present

5. WET TEST FOR CH_3COO^-

Test	Observation	Inference
O.S. + FeCl_3 solution	No change	CH_3COO^- absent

6. WET TEST FOR $\text{C}_2\text{O}_4^{2-}$

Test	Observation	Inference
O.S. + dil. Acetic acid + CaCl_2	_____	_____

NO (Nitrate)

C.T. for

Test	Observation	Inference
1. 0.5 + diphenylamine + conc. H_2SO_4	Intense blue colour	NO_3^- confirmed
2. 0.5 + conc. H_2SO_4 heat and cool add freshly prepared FeSO_4 along side of test tube.	Brown ring forms at the junction of 2 liquids	NO_3^- confirmed

Result:- Given inorganic salt no. 5 contains following

- Basic radical Ba^{2+} (Barium ion)
- Acidic radical..... NO (Nitrate ion)
- Molecular formula of the salt..... $\text{Ba}(\text{NO}_3)_2$
- Hence the name of the salt..... Barium nitrate

Remark and sign of teacher:

NOTES /CALCULATIONS /ROUGH WORK